

MedAssist AI – Medical Symptom Analysis & Disease Prediction System

Milestone 1 Progress Report

1. Project Details

Project Title: MedAssist AI – Medical Symptom Analysis & Disease Prediction System

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Internship: Infosys Springboard Internship

Date: 11 July 2026

2. Objective

Develop an AI-powered healthcare application that predicts diseases from user symptoms, assesses patient health risk, and provides treatment recommendations using Machine Learning and FastAPI.

3. Work Completed

Dataset Collection: Collected and organized Training.csv, Testing.csv, Symptom-severity.csv, symptom_Description.csv, symptom_precaution.csv, medications.csv, diets.csv, workout_df.csv, disease_severity.csv and precautions_df.csv.

Data Preprocessing: Completed data cleaning, missing value handling, label encoding, dataset preparation and model training preparation.

Machine Learning Model: Successfully trained the Disease Prediction Model and generated disease_prediction_model.pkl and label_encoder.pkl.

Backend Development: Developed the FastAPI backend implementing Disease Prediction, Risk Assessment and Treatment Recommendation modules.

API Development & Testing: Developed REST APIs and verified Disease Prediction, Risk Assessment and Recommendation APIs using Swagger UI.

Version Control: Completed Git repository setup, branch creation, commit and push. Successfully uploaded backend implementation to GitHub.

4. Technologies Used

Python, FastAPI, Machine Learning, Scikit-learn, Pandas, NumPy, Uvicorn, Git and GitHub.

5. Current Project Structure

backend/ routes/ predict.py recommendation.py risk_assessment.py schemas.py train_model.py
datasets/ Training.csv Testing.csv symptom datasets medications diets

6. Current Status

Module	Status
Dataset Collection	Completed
Data Cleaning	Completed
Disease Prediction	Completed
Risk Assessment	Completed
Recommendation Module	Completed
FastAPI Backend	Completed
API Testing	Completed
GitHub Push	Completed

7. Challenges Faced

Organizing multiple healthcare datasets, preparing data for machine learning, integrating backend modules and managing Git version control.

8. Next Milestone

Patient Registration Module, User Authentication, PostgreSQL Integration, React Frontend, Backend–Frontend Integration and Dashboard Development.

9. Conclusion

The first milestone has been completed successfully. The backend APIs for disease prediction, risk assessment and treatment recommendation have been developed, tested and pushed to GitHub. The project is now ready for database integration and frontend development.